

Risk Management Task

Category	Identified Risks
Data & Information Risks	1. Delayed or restricted access to Riot API/replay data. 2. Poor physiological data quality (noisy heart rate/eye tracking). 3. GDPR non-compliance leading to delays in approval.
Stakeholder & Communication Risks	4. Limited availability of coaches or players for scrim participation. 5. Miscommunication about project scope or deliverables. 6. Changing client expectations mid-project (new KPIs, extra features).
Methodology & Technical Risks	7. Synchronization of multimodal data fails to achieve <10 ms accuracy. 8. Dashboard usability lower than expected (coaches find it confusing). 9. Technical stack limitations (server capacity, API rate limits).
Timeline & Resource Risks	10. Scope creep from adding more KPIs/modalities than feasible. 11. Team member absence due to illness or workload. 12. Unexpected delays in setting up sensors or capture environment.
External Environment Risks	13. Riot Games API policy changes that break data access. 14. Tournament schedule conflicts reduce scrim opportunities. 15. Changes in league rules restricting physiological device use.

ID	Description	Category	Probability (P)	Impact (I)	Score (P×I)	Priority	Response Strategy
R001	Riot API or replay data delayed/unavailable	Data	2 (Medium)	3 (High)	6	Critical	Mitigation: Cache scrim data locally; use replay parsing tools as backup.
R002	Physiological data too noisy to be usable	Data	2 (Medium)	2 (Medium)	4	Moderate	Mitigation: Run pilot sessions early, test multiple devices, collect redundant signals.
R003	GDPR approval or consent process delays project start	Data	1 (Low)	3 (High)	3	Moderate	Avoidance: Draft consent forms and data protection plan by Week 3; pre-approve with BUas ethics team.
R004	Coaches/players unavailable for feedback or scrims	Stakeholder	3 (High)	2 (Medium)	6	Critical	Mitigation: Schedule sessions well in advance, offer async video demo for feedback.
R005	Misalignment on deliverables (scope misunderstanding)	Stakeholder	2 (Medium)	3 (High)	6	Critical	Avoidance: Share scoping doc for sign-off early, hold mid-project review to confirm alignment.
R006	Client changes KPI requirements late	Stakeholder	2 (Medium)	2 (Medium)	4	Moderate	Contingency: Freeze KPI list by Week 4; defer new requests to future phase.
R007	Synchronization fails (<10 ms not met)	Technical	2 (Medium)	3 (High)	6	Critical	Mitigation: Build early pipeline prototype by Week 5; use relative timestamps if absolute fails.
R008	Dashboard usability too low for adoption	Technical	2 (Medium)	3 (High)	6	Critical	Mitigation: Run quick coach usability test after v1 release, iterate based on feedback.
R009	API rate limits throttle data collection	Technical	2 (Medium)	2 (Medium)	4	Moderate	Mitigation: Throttle requests, use local replay data to avoid over-reliance on live API.
R010	Scope creep adds more KPIs/modalities	Timeline	3 (High)	3 (High)	9	Critical	Avoidance: Strict change control; new requests go to backlog for next sprint.
R011	Team member becomes unavailable	Timeline	2 (Medium)	2 (Medium)	4	Moderate	Mitigation: Cross-train team members, keep shared documentation.
R012	Sensor setup delays eat into data collection time	Timeline	2 (Medium)	2 (Medium)	4	Moderate	Contingency: Schedule buffer sessions, prioritize non-physiology data first.
R013	Riot API policy change removes needed endpoints	External	1 (Low)	3 (High)	3	Moderate	Monitoring: Subscribe to Riot Dev updates, maintain backup replay parsing strategy.
R014	Tournament schedule conflicts with planned testing	External	2 (Medium)	2 (Medium)	4	Moderate	Mitigation: Plan data collection outside tournament windows, schedule buffer weeks.
R015	League bans physiological sensors	External	1 (Low)	3 (High)	3	Moderate	Contingency: Focus on telemetry + comms analysis, skip physiology if prohibited.

Integration Plan

1. Pre-Green Light Risk Actions (Weeks 3 - 4)

- **Finalize Scope & KPI List:** Lock in a realistic set of KPIs this week to prevent scope creep (R010). Present these to the client during the green light session for approval.
- **Pilot Capture & Data Quality Check:** Run a quick capture session before the green light presentation to validate data pipeline synchronization (R002, R007). Share early results as proof-of-feasibility.
- **Draft GDPR Documentation:** Have consent forms, privacy notice, and data handling plan ready to present (R003). Getting early sign-off reduces risk of compliance-related delays.
- **Align on Stakeholder Availability:** Confirm player scrim schedule and coach feedback sessions for Weeks 5–8 (R004). Present this plan at the green light to secure commitment.

2. Immediate Post-Green Light (Weeks 4 - 5)

- **Incorporate Client Feedback:** Update scope or KPI definitions only if client explicitly requests changes during the green light session (R005, R006).
- **Kick Off Usability Planning:** Schedule first coach usability session for dashboard v1 in Week 6 to de-risk adoption (R008).
- **Confirm Tournament Dates:** Lock down testing sessions to avoid conflicts (R014).

3. Ongoing Monitoring & Reporting (Weeks 5 - 8)

- **Weekly Risk Review:** Include risk register updates in weekly stand-ups. Focus on critical risks (R001, R004, R007, R008, R010).
- **Escalation Triggers:** Any risk score ≥ 6 flagged for mentor review immediately; if blocking, escalate to Bram Heijligers and client contact.
- **Update Risk Register:** After first coach feedback session, re-evaluate probabilities for adoption/usability risks (R008).

4. Contingency Planning & Buffers

- **Technical Fallbacks:** If physiology data proves unreliable (R002), switch to telemetry + video-only insights and document this.
- **Change Control:** New client requests captured but deferred to backlog after Week 4 unless they can be delivered without jeopardizing MVP.
- **Time Buffers:** Maintain at least 1 week at the end of the block reserved for integration, testing, and presentation polish.

5. Impact on Project Deliverables

- **Green Light Presentation:** Risk register summary included as a slide to demonstrate readiness.
- **Timeline Section in Proposal:** Updated to reflect that risk management actions (pilot, scope lock, consent forms) are already in motion.
- **Confidence Building:** Showing early data capture results + consent workflow will reassure collaborators that technical, ethical, and scheduling risks are under control.